

Impact of Adding, Removing and Modifying Driving and Non-Driving Related Information on Trust in Autonomous Vehicles

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ABSTRACT

Integrating Autonomous Vehicles (AVs) into daily life requires a fundamental shift in user trust and acceptance. Previous studies show that Augmented Reality (AR) visualization can improve trust in AVs. However, those studies primarily focus on adding information related to the driving task. The question then must be asked: What happens if we remove unnecessary information or modify information unrelated to the driving task? Our work presents six AR visualizations within AVs, including adding, modifying, and removing information related to or non-related to the driving task. We evaluate the influence of these visualizations on trust, technology acceptance, and Situational Awareness (SA) in AVs by conducting a study employing a custom-developed driving simulator. Results show that using AR to add, modify, or remove information can significantly increase trust in AVs.

Index Terms: Trust, Visualization design and evaluation methods, Augmented Reality, Virtual reality, Autonomous vehicle.

1 INTRODUCTION

Autonomous vehicles (AVs) have the potential to improve road safety and reduce congestion but face trust challenges due to safety concerns in complex scenarios, such as poor weather or pedestrian interactions [3, 8]. Enhancing trust in AVs is crucial for user acceptance [9]. Augmented Reality (AR) allows real-time visualization of AV decisions, but prior research largely focuses on adding information related to the driving task, leaving gaps in understanding the role of modifying or removing information, particularly for non-driving related information. This study addresses these gaps by testing six AR visualizations based on two dimensions: information type (related/non-related to driving tasks) and AR forms (adding, modifying, removing). We distinguish between two types of information: those related and non-related to the driving task. Information related to driving tasks includes details about the vehicle, dynamic objects like pedestrians and other vehicles, and static objects such as traffic signs. The term non-related to the driving task refers to any information unrelated to driving or traffic laws, such as points of interest (PoIs) or the vehicle's aesthetics. A Virtual Reality (VR) driving simulator was used to evaluate these visualizations' impact on trust, situational awareness (SA), and acceptance.

2 METHODOLOGY

This experiment aimed to answer our research question – *How will AR affect trust, SA, and User Acceptance if we modify, add, or remove the information in the environment through AR visualization?*

Simulation Environment Set Up: Based on our research question, we developed seven conditions (one for each of the six AR visualizations and one baseline condition with no AR), all set within

an urban environment. All seven conditions were performed within a custom VR driving simulator to simulate the AR visualization and provide a variety of road types, including tunnels, left-turn-only lanes, inclines, and declines; the simulator includes elements like traffic lights, pedestrians, and other vehicles. These paths were randomized between the different visualization conditions. Each path contains junctions where (1) pedestrians suddenly cross the street, causing the AV to decelerate and stop to wait for them; and (2) other vehicles approach from the left, right, or both sides without stopping, leading the AV to brake and reduce speed. The vehicle's speed was set to 40 km/h and decelerated to 30 km/h for sharp turns. This project uses the open source application for simulation AVs in Unity3D: AutoWSD [7]. An Oculus Quest Pro head-mounted display and Unity version 2020.3.22f1 were used. The experiment was conducted in a quiet room with seated participants.

User study: We recruited 28 participants in total. Participants were students or members of the laboratory, with 18 males and 10 females aged 22 to 50 (mean=28.18, SD=7.8). They were not paid for their participation. On a 5-point Likert scale (1 = Strongly Disagree — 5 = Strongly Agree), participants showed medium trust in AVs (M=3, SD=0.798), and 75% of the participants hold a driver's license. The experiment started with an introduction to the experiment. Participants then filled out a demographic questionnaire (age, gender, trust in AV, and driver's license). The six AR visualizations were explained to the participants. They were asked to sit in a chair and don the Oculus Quest Pro headset. Participants were given a short AV trip with the baseline condition. The participants were then presented with the seven conditions in a random order. After each trip, participants completed a questionnaire assessing trust, SA, user acceptance, and likability. At the end of the experiment, they participated in a short interview and provided general feedback. The study had approval of the university ethics committee.

AR Visualization Conditions: Add Related: Displays navigation paths, alerts for incoming vehicles, and notifications about lane changes. Add Non-Related: Introduces PoIs, such as cafes, landmarks, and dinosaur parks. Modify Related: Highlights dynamic objects, such as pedestrians suddenly crossing the street. Modify Non-Related: Enhances interior aesthetics (e.g., futuristic decor). Remove Related: Remove irrelevant pedestrians to reduce clutter. Remove Non-Related: Eliminates blind spots by making car elements transparent (e.g., roof, doors).

Measure: To examine if adding, modifying, or deleting information in the environment can increase trust in AVs, we analyzed the following qualitative data: Trust in Automation questionnaire by Körber [4], the Situation Awareness Rating Technique (SART) [2], Likability [1], Technology Acceptance Model (TAM) [5].

3 RESULTS

The parametric data is analyzed using repeated measures ANOVA to compare the seven conditions. The normality assumption is checked by computing the Shapiro-Wilk test. We performed multiple pairwise t-tests, and p-values were adjusted using Bonferroni correction. For non-parametric data, Friedman's ANOVA test was

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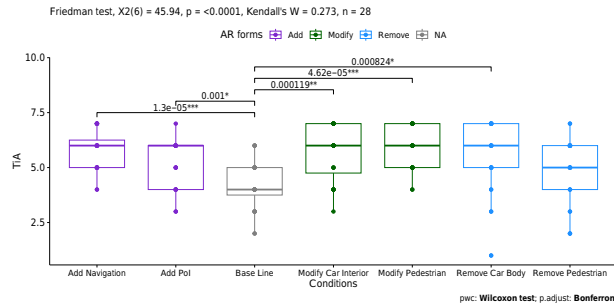


Figure 1: Results of Friedman's ANOVA test for trust.

used. Posthoc analysis was performed using a paired Wilcoxon signed-rank test, and p-values were adjusted using the Bonferroni correction. We utilized the ggboxplot function [6] to create the figure, which features a box plot illustrating data distribution and also provides statistical details, including the test used, chi-square, p-value, effect size, and sample size.

Trust in Automation: Using AR technology for addition, modification, or removal of information (except Removed Pedestrian) demonstrated a statistically significant increase in trust ratings when compared to the baseline condition ($p < 0.0001$), see Figure 1. Adding non-driving related information, such as PoIs, increased trust but resulted in higher risk perception compared to the modified related scenario.

Situational Awareness: Understandable was rated significantly higher when using AR technology to add and remove related or non-related information, or modify information related to the driving task. These ratings were notably higher compared to both baseline and scenario modification information non-related to the driving task. However, when considering the Supply factor, no statistically significant differences were observed between the baseline scenario and the other scenarios. Supply received significantly higher ratings in other scenarios compared to the modified non-related scenario (the modification of the interior car),

Technology Acceptance: Perceived Usefulness (PU) was highest for driving-related additions and modifications (Repeated Measures ANOVA, $F(6,162) = 28.16$, $p < 0.0001$). Perceived Ease of Use (PEOU) was also positively influenced by driving-related AR strategies ($F(1.51,40.8) = 10.1$, $p = 0.0008$).

Likability: AR conditions involving non-driving-related modifications (e.g., aesthetic enhancements) received high likability scores. Participants expressed a preference for customizable AR interfaces tailored to their preferences.

4 DISCUSSION

This study highlights the potential of AR visualizations in addressing AV adoption challenges. Specifically, it shows how AR designs can improve transparency, match user preferences, and balance functional utility with aesthetic appearance. Enhancing transparency involves adding and modifying driving-related information, such as navigation paths and object detection, which can improve user understanding and confidence. These visualizations effectively bridge the gap between AV actions and passenger perceptions, especially in uncertain environments. The ability to personalize AR visualizations is critical for user satisfaction and trust. Participants expressed a strong interest in features like personalized navigation feedback, customization of the interior, and selective overlays for PoIs. Personalization ensures that the visualizations resonate with individual user needs and adapt to diverse situational contexts, making the AV experience more engaging and user-friendly. Aesthetic features like interior designs can increase trust and likability, but they sometimes distract users from essential driving-related information. This shows the need to design AR

visuals that fit the context, ensuring that aesthetic elements complement rather than detract from functional utility. Balancing usefulness and aesthetics is important for improving the passenger experience.

5 CONCLUSION

In this paper, we proposed six AR visualizations to improve trust in AVs. We classified these visualizations into three forms of AR (add, remove, and modify) and into two categories of information (related and non-related to the driving task). Throughout a series of different conditions, we compare the impact of these visualizations on trust, TAM, SA, and Likability. Driving-related additions and modifications were particularly effective, offering essential transparency and clarity during uncertain scenarios. Non-driving-related enhancements improved trust and user enjoyment, though they require careful integration to avoid distraction.

Future Directions: Future research should explore long-term trust development through extended real-world testing. Incorporating diverse demographic groups will help generalize findings and identify varying user needs. Addressing the ecological validity of VR simulations by integrating physical motion and higher fidelity environments will further refine AR visualization designs. The challenge lies in determining the optimal amount of information and when to present it to mitigate the user's mental workload.

Design Recommendations: Avoid removing dynamic elements, such as pedestrians, to ensure safety. Non-driving-related AR content should be limited in risk scenarios to minimize distractions. Providing robust customization options ensures the visualizations cater to diverse user preferences, enhancing engagement and satisfaction.

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